TECHNICAL INTERNSHIP PRESENTATION

Vaanmegam Internship

VERA-TDMA: Centralized TDMA Topology Optimization & Independent Verification System

---

PRESENTER

Saumiya Varalakshmi E

DOMAIN

Wireless Networks & Graph Optimization

VERA-TDMA | CENTRALIZED TOPOLOGY OPTIMIZATION & VERIFICATION

1 INPUT & TOPOLOGY — Node Coordinates → Communication Graph

Input: Wireless Node Coordinates

Each node is represented by Cartesian coordinates (metres).

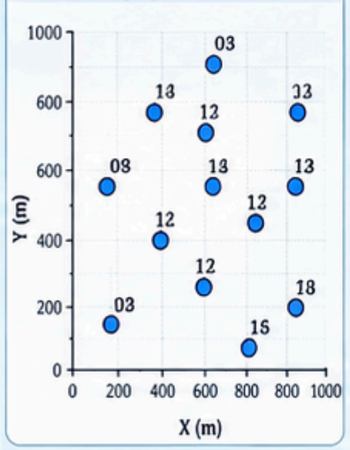
$$\text{Node}_i = (x_i, y_i)$$

Number of nodes ($|V|$) = 16
Communication range (R) = 500 m

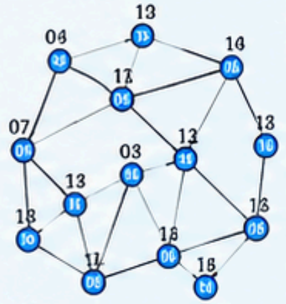
Node ID	x (m)	y (m)
Node_01	120	340
Node_02	450	220
Node_03	780	610
...	...	...
Node_16	690	180

Input validation:
Checks for unique node IDs and valid coordinates.

Node Deployment (16 Nodes)



Communication Graph G = (V, E)



Two nodes are directly connected if:

$$d(i,j) = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \le R$$

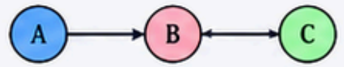
(Uses squared distance: $(x_i - x_j)^2 + (y_i - y_j)^2 \le 500^2$)

|V| = 16	|E| = 42

The system identifies 42 direct communication links.

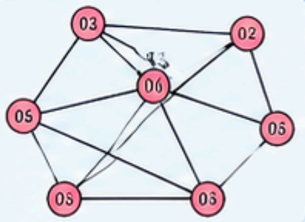
2 DISTANCE-2 CONFLICT MODEL — Direct Links + Two-Hop Conflicts

Two-Hop Interference via Common Neighbours



A and C are two hops apart.
Under the Distance-2 model,
they are treated as a scheduling conflict.

Conflict Graph $G_c = (V, E_c)$



Conflict Constraints

For every conflict edge $(u, v) \in E_c$:

$$\text{slot}(u) \neq \text{slot}(v)$$

Conflict Graph Details

Direct communication links	= 42
Distance-2 conflict edges	= 90
Total conflict edges ($\|E_c\|$)	= 90

The Distance-2 model is a protocol-level abstraction. It does not attempt to model physical RF effects such as SINR, antenna patterns, fading, capture effect, or cumulative interference.

3 INITIAL TDMA SCHEDULING — DSATUR Graph Colouring

DSATUR Scheduling Process

1. Compute saturation degree for each node
2. Select node with highest saturation degree
3. Assign smallest available slot (colour) without conflicts
4. Update saturation degree and repeat

Optimization Objective

minimize K
subject to:
$\text{slot}(u) \neq \text{slot}(v)$ for every $(u, v) \in E_c$

Graph Colouring Illustration



Initial DSATUR Result

Initial frame = 9 slots

K = 9

4 SCHEDULE REFINEMENT — Class Elimination & Kempe-Chain

Class Elimination

Attempt to reduce slots by elimintating complete colour classes. Nodes in the selected slot are reassigned to other slots while maintaining all conflict constraints.

Accepted class eliminations = 0

No slot could be removed through this optimization stage.

Kempe-Chain Refinement

Attempt to restructure assignments involving two slots by exchanging labels of a valid two-colour connected component.

Kempe attempts = 42

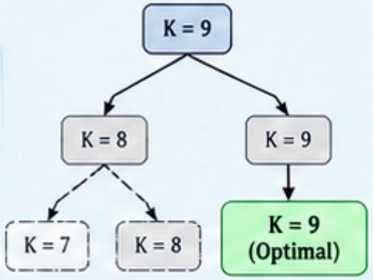
Accepted Kempe moves = 0

No beneficial changes were found.



5 EXACT OPTIMALITY CHECK — DSATUR Branch-and-Bound Solver

Branch-and-Bound Search



Solver Details

Exact DSATUR branch-and-bound solver searches for a solution using fewer slots and determines optimality for this topology.

Results	
Generated frame (K)	9
Exact optimum	9
Solver status	OPTIMAL
Optimality gap	0 slots
Relative gap	0%

The generated 9-slot schedule is exactly optimal for this topology under the implemented Distance-2 model.

6 FINAL SCHEDULE, VERIFICATION & PUBLICATION

Final Node-Slot Mapping (K = 9)

Node ID	Slot	Node ID	Slot
Node_01	0	Node_09	6
Node_02	1	Node_10	7
Node_03	2	Node_11	8
Node_04	0	Node_12	6
Node_05	3	Node_13	0
Node_06	4	Node_14	1
Node_07	5	Node_15	2
Node_08	3	Node_16	0

Independent Verification

Direct conflict checks (42 pairs)	0 violations
Two-hop conflict checks (90 pairs)	0 violations
Total conflict violations	**0**

Verification Result
PASS ✓

Publication Gate
PASS ✓

OVERALL WORKFLOW PIPELINE

16 Nodes (Input) → 42 Direct Links (|E|) → 90 Distance-2 Conflict Edges ($|E_c|$) → 9-Slot Initial Schedule (DSATUR) → Schedule Refinement (Class Elimination & Kempe-Chain) → Exact Optimum K = 9 (Branch-and-Bound) → Optimality Gap 0 slots (0%) → 132 Conflict Checks (42 + 90) → 0 Violations → PASS (Verified & Published)

KEY FEATURES

- ✓ Minimum-slot objective for static topology
- ✓ Spatial reuse for non-conflicting nodes
- ✓ Two-hop conflict awareness (distance-2 model)
- ✓ Schedule refinement (class elimination & Kempe-chain)
- ✓ Exact solver proof for this demonstrated topology
- ✓ Independent verification before publication

TECHNICAL SCOPE

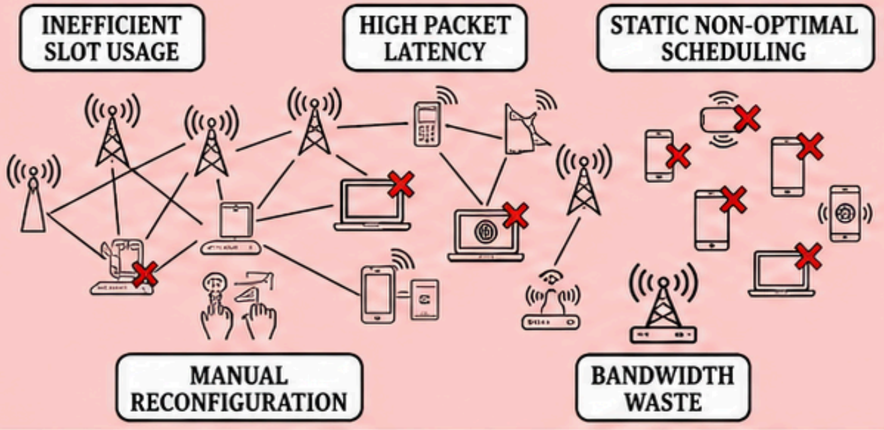
⚠ Optimality is proven for the demonstrated 16-node topology under the implemented distance-2 model.
This is a protocol-level abstraction and does not model physical RF propagation effects such as SINR, antenna patterns, fading, capture effect, or cumulative interference.

VERA TECH

VERA-TDMA CENTRALIZED TOPOLOGY OPTIMIZATION & VERIFICATION (VERA-TDMA)

CHALLENGES in Conventional TDMA Scheduling

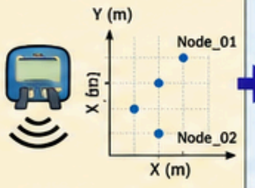
Traditional TDMA scheduling faces complexity in **static wireless network optimization**. Conventional scheduling relies on **fixed, non-optimized slots and manual network configuration**, leading to **wasted network capacity** and increased latency for crucial communications.



PROPOSED SOLUTION (flow cont.)

VERA-TDMA is a centralized system designed to generate a minimum-length, transmission schedule for static wireless networks. It combines **topology construction**, **direct** and **two-hop conflict analysis**, and **graph coloring** to derive an exact optimal schedule. It uses **DSATUR graph coloring with exact solver verification and validation** to adaptively select the minimal slots and slot assignments, enabling efficient, real-time-like and autonomous schedule generation.

NODE COORDINATE ACQUISITION



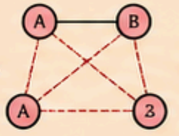
Captures 2D node coordinates.

TOPOLOGY CONSTRUCTION



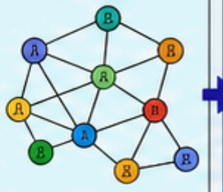
Euclidean distance calculations for links.

CONFLICT GRAPH DERIVATION



Finds direct and two-hop interference pairs.

GRAPH COLORING SCHEDULER



DSATUR priority and slot assignment.

EXACT OPTIMIZATION



Branch-and-bound for exact optimum check.

FINAL TDMA SCHEDULE OUTPUT

Node	Slot	Nodes
Node_01	0	Node_05, Node_04
Node_03	1	Node_01, Node_09
Node_04	3	Node_02, Node_08
Node_05	4	...

Accepted class eliminations = 0
Frame length remains 9 slots
Exactly Optimal : 9 slots

INDEPENDENT VERIFICATION



Checks all conflict constraints (slot(i) ≠ slot(j)).

PUBLICATION GATE



Published only if pass validation.

CONTINUOUS REFINEMENT & OPTIMIZATION LOOP

PROBLEM RESOLUTION

- **Slot Optimization:** Instead of using arbitrary slot counts, VERA-TDMA targets the exact minimum slots.
- **Spatial Reuse:** Enables simultaneous transmissions in non-conflicting nodes.
- **Real-Time Generation:** Optimized pipeline generates schedules quickly for small topologies.
- **Conflict Elimination:** Identifies and eliminates direct and two-hop conflicts.

NOVELTY

Exactly Optimal Solutions
- Finds minimal slots with mathematical proof.

Integrated Two-Hop Model
- Direct and hidden-node interference is natively handled.

Verification and Validation
- Independent post-scheduling checks of integrity.

Pipeline Integrity Assurance
- Validated input and proven optimal output.

KEY FEATURES

- **Exact Optimality:** Determines minimum possible slots for a static topology.
- **Two-Hop Interference Modeling:** Prevents conflicts from hidden nodes.
- **Conflict-Free Schedule:** Ensures reliable, collision-free communication.
- **Centralized Control:** Ideal for small static networks.
- **Optimization Pipeline:** Integrated flow from coordinates to optimal schedule.

IMPACT AND BENEFITS

VERA-TDMA: Centralized TDMA Topology Optimization & Independent Verification

IMPACT

Eliminates transmission collisions
Generates conflict-free TDMA schedules using direct and two-hop interference modelling.

Maximizes spectrum utilization
Produces minimum-length schedules with mathematical optimality verification.

Enables scalable network planning
Works for arbitrary static topologies with deterministic and repeatable results.

Reduces planning time and manual effort
Automates topology construction, scheduling, optimization and validation.

Supports reliable and efficient communication
Ensures interference-free transmissions, improving network performance and reliability.

GOVERNMENT / STRATEGIC ALIGNMENT

DoT : Efficient and reliable spectrum usage for wireless networks

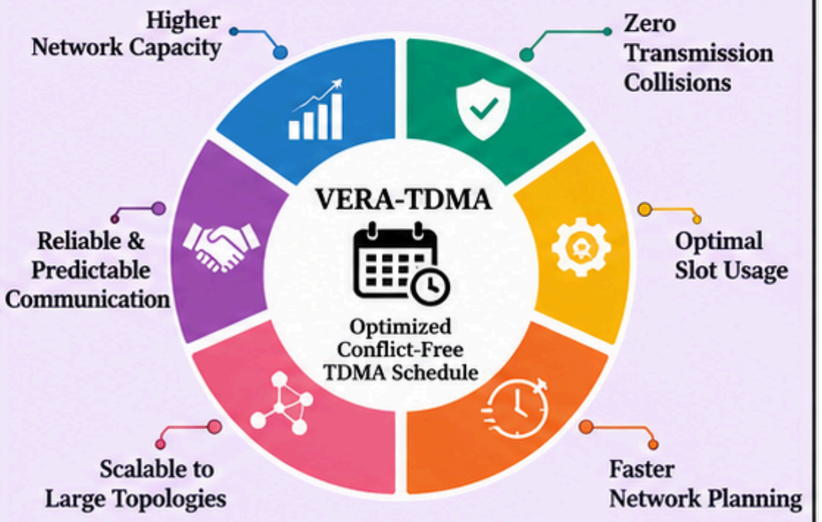
MeitY : Promoting indigenous algorithm development and network optimization tools

Indian Armed Forces : Reliable communication in static tactical networks (base camps, radar, UAVs)

DRDO : Support for secure and interference-free communication systems

Make in India : Strengthening indigenous R&D in wireless communication and optimization technologies

KEY IMPACT AREAS



BENEFITS AFTER DEPLOYMENT

Efficient Spectrum Usage
- Minimum-length TDMA schedule
- Maximizes channel utilization

Reduced Operational Overhead
- Automated schedule generation
- Minimizes manual configuration and testing time

Conflict-Free Communication
- Eliminates direct and two-hop conflicts
- Improves reliability and throughput

Easy Network Expansion
- Handles new nodes with minimal reconfiguration
- Deterministic and repeatable results

BENEFITS TO STAKEHOLDERS

Government Agencies	Network Operators	Industry & Enterprises	Research Community	Academic Institutions
Efficient spectrum management and policy support	Higher capacity and reliable network operation	Optimized wireless deployment and lower operational cost	Benchmarking dataset and reproducible tools	Supports education, research and real-world applications

SUSTAINABLE DEVELOPMENT GOALS (SDGs)



LONG-TERM IMPACT

Foundation for advanced MAC protocol research	Enables next-generation cognitive and adaptive TDMA systems	Supports indigenous development of network optimization tools	Applicable to defence, industrial and rural connectivity networks

RESEARCH AND REFERENCES

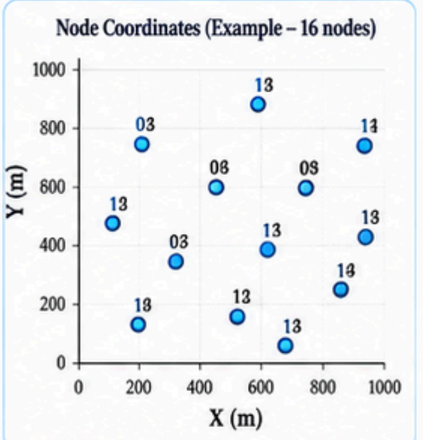
VERA-TDMA: Centralized TDMA Topology Optimization & Independent Verification

Research Gap: Existing TDMA scheduling methods either rely on fixed heuristics, ignore two-hop interference, or lack a unified framework that combines topology construction, optimization, exact verification, and independent validation for static wireless networks.

KEY RESEARCH PAPERS & REFERENCES

1. Brélaz, D. (1979) "New Methods to Color the Vertices of a Graph"	DSATUR graph colouring algorithm for optimal/near-optimal scheduling.
2. Garey, M. R., & Johnson, D. S. (1979) "Computers and Intractability: A Guide to the Theory of NP-Completeness"	Graph colouring and scheduling complexity (NP-hard nature).
3. Burton, D., & Tsang, E. (2000) "Bandwidth Allocation in Wireless Mobile Networks"	TDMA scheduling and interference modelling in wireless networks.
4. Ramanathan, R., & Rosales-Hain, R. (2000) "Topology Control of Multihop Wireless Networks Using Transmit Power Adjustment"	Topology construction and interference relationships.
5. Li, X. et al. (2016) "Two-Hop Interference Aware Scheduling in Wireless Networks"	Importance of two-hop conflict modelling for practical TDMA.
6. Cormen, T. H. et al. (2009) "Introduction to Algorithms"	Graph algorithms, branch-and-bound optimization techniques.
7. Recent Advances (2020–2024) Adaptive and learning-based MAC scheduling	Inspiration for future enhancements (e.g., RL-based adaptive scheduling).

DATASET / INPUT SOURCES



Input Parameters

- Node IDs
- Cartesian coordinates (x, y)
- Communication range (R)
- Static network topology

Derived Data

- Communication graph $G = (V, E)$
- Distance-2 conflict graph $G_c = (V, E_c)$
- Conflict constraints for scheduling

APPLICATIONS & MARKET RELEVANCE

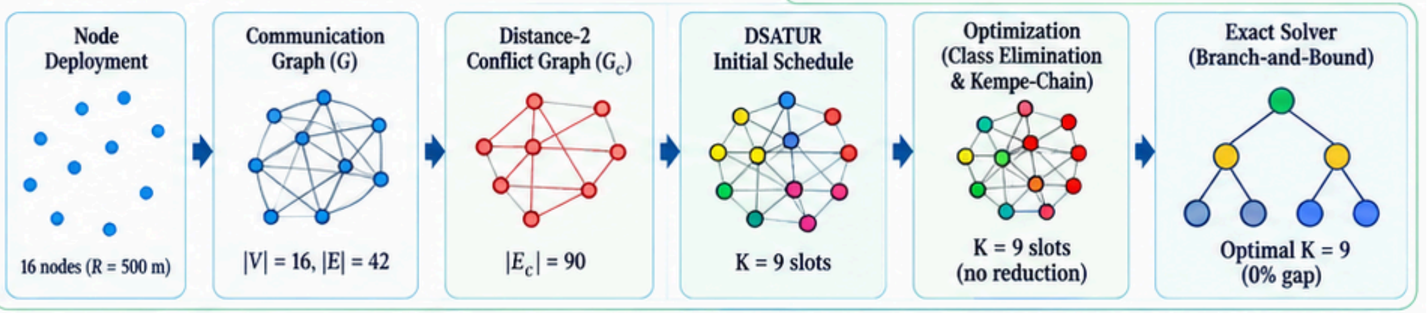
Tactical & Defence Communications
Reliable, interference-free spectrum usage for static deployments (base camps, radars, UAVs).

UAV and Swarm Networks
Optimized channel allocation for multi-UAV coordination.

Industrial IoT & Private Networks
Efficient spectrum sharing in smart factories and critical infrastructure.

Rural & Remote Deployments
Improved spectrum utilization in limited-spectrum environments.

SIMULATION / METHODOLOGY REFERENCES



TOOLS & TECHNOLOGIES

- **Python** (NetworkX, NumPy, Pandas)
- **Matploltib** Visualization & Analysis
- **Gurobi / OR-Tools** Exact Solver
- **Jupyter** Development Environment
- **NetworkX** Graph Modelling
- **Custom Simulator** Topology & Scheduling Pipeline